IP/WP Cover Sheet

MBA 2024/2025

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Individual Project Report

1. Individual Project Brief

My placement at the Cambridge Centre for Alternative Finance (CCAF) focused on developing comprehensive methodologies for analysing energy consumption patterns in generative AI (GenAI) inference operations. This work is particularly timely, given that inference operations are projected to account for up to 90% of total AI system energy consumption (Luccioni, 2025) and could represent a staggering £50-80 billion in annual global costs by 2027. Alternative finance in the UK encompasses innovative financing mechanisms that operate outside traditional banking systems, including peer-to-peer lending, crowdfunding platforms, cryptocurrency exchanges, digital investment services, and equity crowdfunding (Cambridge Centre for Alternative Finance, 2025). These mechanisms have transformed the financial landscape by providing direct connections between borrowers and lenders, and bypassing conventional financial intermediaries.

CCAF positions itself as the leading global research institution providing evidence-based insights to policymakers, industry practitioners, and academic communities on emerging financial technologies. The centre operates at the intersection of finance, technology, and policy, conducting rigorous empirical research that influences regulatory frameworks, industry standards, and strategic decision-making across the alternative finance ecosystem.

The centre established significant credibility through rigorous analysis of disruptive financial technologies, particularly demonstrated through their comprehensive cryptocurrency energy consumption assessment project. This previous work provided invaluable methodological foundations for understanding complex, distributed technological systems whilst addressing stakeholder concerns around economic implications, environmental sustainability, and regulatory compliance requirements.

Building upon this methodological foundation, my role as a GenAI analyst involved extending these analytical capabilities to address the rapidly growing energy demands of AI inference systems. The project represented a natural evolution of CCAF's research agenda, recognising that alternative finance platforms increasingly depend on AI technologies for credit scoring, fraud detection, automated trading, and risk assessment.

My primary responsibilities centred on quantifying the relationship between model complexity, infrastructure deployment patterns, and energy consumption across major cloud providers and specialised AI hosts. This involved developing sophisticated analytical frameworks that could account for technical nuances of different AI architectures, efficiency characteristics of various hardware

configurations, and economic implications of deployment strategies across different provider ecosystems.

The project aimed to establish robust measurement frameworks addressing three critical areas: cost modelling for AI deployment strategies, regulatory compliance with emerging sustainability reporting requirements, and competitive positioning analysis for financial services firms evaluating AI infrastructure options.

The economic significance of GenAI inference energy consumption extends beyond environmental considerations to fundamental questions of operational scalability and cost structure. Current estimates suggest that inference operations represent 80-90% of total AI system energy consumption over typical deployment lifecycles, with individual queries ranging from £0.0001 to £0.02 in energy costs depending on model complexity and infrastructure efficiency (Amazon Web Services, 2024). For alternative finance platforms processing millions of daily transactions through AI-powered systems, these costs aggregate to material operational expenses that directly impact profitability and competitive positioning.

2. Individual Project Context

CCAF operates within the rapidly evolving landscape of sustainable finance and emerging technology assessment, leveraging its unique position within Cambridge Judge Business School to provide access to both academic rigour and extensive industry partnerships. This positioning enables research that influences policy development, regulatory frameworks, and industry best practices across alternative finance mechanisms whilst maintaining analytical independence (Cambridge Centre for Alternative Finance, 2025).

The cryptocurrency energy analysis project established crucial methodological precedent for approaching complex technological energy assessment within financial services contexts. This prior work demonstrated CCAF's capability to quantify energy consumption in complex, distributed systems whilst addressing economic implications, regulatory compliance requirements, and policy considerations that directly affect industry development. The success of this analysis provided substantial credibility with industry partners whilst establishing proven frameworks for handling uncertain data sources and presenting sophisticated technical analysis to diverse stakeholders.

My GenAI energy analysis project addresses a particularly timely intersection for CCAF's mission, focusing on emerging challenges of AI inference energy consumption that increasingly dominate operational costs for AI-dependent organisations across the alternative finance spectrum. Financial institutions face growing regulatory pressure to report Scope 3 emissions from cloud computing services whilst simultaneously seeking to maintain competitive AI capabilities for customer acquisition, risk management, and operational efficiency.

Traditional measurement approaches relied heavily on crude estimation methods that failed to capture nuanced efficiency differences across providers, deployment strategies, and geographic regions. These limitations created significant gaps in understanding the true cost and environmental implications of AI adoption, making informed strategic decision-making difficult for organisations evaluating different technology procurement options.

From an economic perspective, GenAI inference energy consumption represents a rapidly growing cost centre for alternative finance organisations. Industry analysis suggests that inference energy costs could reach 15-25% of total cloud computing expenses for AI-intensive organisations by 2026, making energy efficiency a critical factor in technology procurement decisions (Meta Platforms Inc., 2024). This economic reality drives substantial demand for sophisticated measurement frameworks that can inform both cost optimisation initiatives and sustainability reporting requirements.

The alternative finance sector presents particularly compelling use cases for sophisticated AI energy analysis. Peer-to-peer lending platforms depend heavily on AI for automated credit scoring and risk assessment, often processing millions of loan applications through complex machine learning models. Cryptocurrency exchanges utilise AI for real-time fraud detection, market analysis, and trading optimisation, creating continuous inference workloads. Digital investment platforms increasingly rely on AI for portfolio optimisation and personalised financial advice.

My role addressed this gap by developing evidence-based methodologies that could quantify energy consumption patterns with sufficient precision to support informed procurement decisions and regulatory reporting. The work required understanding both technical characteristics of AI infrastructure and business contexts within which financial institutions operate these systems, particularly focusing on economic implications of different deployment strategies.

The economic context proves particularly relevant as energy costs continue rising globally whilst AI inference demands scale exponentially. Current projections suggest that AI inference energy consumption could represent £50-80 billion in annual global costs by 2027, making efficiency optimisation a critical competitive advantage (Equinix Inc., 2024).

3. Goals and Objectives

I approached this placement with carefully defined learning objectives aligned with my MBA programme goals and career aspirations in technology strategy, sustainable finance, and strategic consulting. My primary objective focused on developing advanced expertise in handling complex, multi-source datasets whilst maintaining analytical rigour in emerging technology assessment contexts. This capability development directly supported my career goals in strategic consulting and technology investment analysis roles post-MBA.

The interdisciplinary nature of the project provided exceptional opportunities to strengthen skills across multiple domains: quantitative data analysis, AI systems understanding, financial sector applications, and policy implications. I aimed to develop sophisticated analytical capabilities that could bridge technical depth with business relevance, aligned with the MBA programme's emphasis on developing strategic thinking capabilities for technology-intensive business environments.

My technical learning objectives encompassed developing proficiency in advanced data collection methodologies, statistical analysis frameworks, and uncertainty quantification approaches suitable for emerging technology assessment. I sought to master synthesising information from diverse sources including financial disclosures, technical specifications, regulatory filings, and industry reports whilst building upon methodological foundations established through CCAF's cryptocurrency work.

Advanced data visualisation and communication capabilities represented another critical development area aligned with my career objectives in consulting roles. I aimed to master sophisticated dashboard development and interactive analysis tools whilst ensuring accessibility for diverse stakeholder groups. These skills complement the MBA programme's emphasis on effective communication and stakeholder management whilst building technical capabilities valuable in technology-focused advisory roles.

I also established learning objectives around developing deep understanding of AI infrastructure economics from both cost and environmental perspectives, recognising this as crucial insight into one of the most significant technological trends affecting business strategy. The project offered opportunities to connect technical considerations to broader business strategy frameworks and financial planning processes.

Personal development objectives included enhancing my ability to work independently on complex analytical projects whilst effectively managing stakeholder relationships and communicating technical findings to diverse audiences. I recognised this placement as an opportunity to bridge theoretical MBA learning with practical application in an emerging technology domain with significant economic implications.

4. Individual Project Experience

4.1 Initial Challenges and Methodology Development

The project commenced with fundamental challenges around data availability and methodological consistency that required innovative solutions. Traditional approaches to measuring AI infrastructure energy consumption relied heavily on capital expenditure proxies, using global conversion factors of approximately £8 million per MW to estimate capacity from financial disclosures. While this

approach offered straightforward implementation, it suffered from systematic biases across different provider types and failed to capture nuanced efficiency differences that directly impact costs and environmental performance.

My initial weeks involved extensive literature review across academic publications, industry reports, regulatory filings, and technical documentation to understand existing measurement approaches and identify limitations. This comprehensive analysis revealed significant gaps in current methodologies, particularly around handling providers with different disclosure practices and accounting for regional variations in infrastructure costs and efficiency characteristics.

Developing the evidence-led methodology framework represented the project's most significant analytical challenge, requiring careful balance between analytical rigour and practical applicability given real-world data constraints. I established a sophisticated three-track classification system: Track A for organisations providing direct capacity disclosures, Track B for organisations requiring proxy indicators, and Track C for organisations necessitating financial estimates and modelling approaches.

This classification system required establishing detailed criteria for evidence quality assessment whilst maintaining analytical comparability across providers with vastly different transparency practices. I needed to develop robust approaches for handling uncertainty, validating assumptions, and communicating confidence levels that would enable stakeholders to make informed decisions based on analytical outputs.

The economic implications of GenAI inference energy consumption became increasingly apparent throughout methodology development. Unlike cryptocurrency mining, which represents discrete computational processes, AI inference involves continuous operational workloads with varying intensity and complexity that change based on query types, model architectures, and infrastructure optimisation strategies.

4.2 Data Collection and Integration

I systematically collected and validated data from over 50 diverse sources, presenting substantial logistical and analytical challenges. Sources ranged from SEC filings and quarterly earnings reports to sustainability disclosures, industry analyses, and technical specifications from equipment manufacturers. Each source type required different collection approaches and validation methodologies whilst presenting unique challenges around data consistency and accuracy.

Financial disclosure analysis proved particularly complex due to significant variations in reporting standards across providers and jurisdictions. Microsoft's comprehensive disclosure practices provided high-quality inputs for

methodology validation, including detailed breakdown of AI-specific capital expenditure and efficiency optimisation initiatives (Microsoft Corporation, 2024). However, other major providers required developing sophisticated inference methodologies that could extract relevant insights from limited disclosure whilst maintaining analytical rigour.

I developed systematic approaches for handling these variations whilst maintaining transparent uncertainty quantification and clear communication about confidence levels. This required creating detailed documentation protocols, validation frameworks, and quality assurance processes that could ensure analytical consistency whilst accommodating diverse characteristics of different information sources.

Technical specifications from GPU manufacturers and data centre operators required careful validation against real-world deployment patterns. Theoretical design parameters often differ significantly from operational performance, particularly around power consumption under varying utilisation levels and cooling requirements. I developed comprehensive cross-referencing approaches that compared manufacturer specifications with disclosed operational metrics where available.

Regional cost factor development proved more analytically complex than initially anticipated, requiring sophisticated understanding of local construction markets, energy infrastructure availability, and regulatory requirements. Data centre construction costs vary dramatically based on local labour markets, land availability, energy costs, and cooling requirements. Nordic regions demonstrate substantially lower construction costs per MW due to favourable cooling conditions and abundant renewable energy availability, whilst tropical and high-cost urban regions require significant adjustments to baseline conversion factors (Google LLC, 2025).

4.3 Technical Analysis and Infrastructure Assessment

The technical analysis component required developing detailed understanding of AI hardware characteristics, data centre cooling systems, power distribution infrastructure, and complex interactions between these components that determine overall system efficiency. This technical foundation proved essential for accurate analysis and credible stakeholder communication whilst enabling nuanced assessment of efficiency claims and identification of optimisation opportunities.

Understanding AI hardware evolution, particularly the rapid transition from H100 to Blackwell GPU architectures and development of specialised AI inference chips, created ongoing challenges around maintaining current technical assumptions whilst projecting future efficiency trends. I developed systematic approaches for monitoring hardware development announcements whilst

maintaining analytical frameworks that could adapt to technological evolution without compromising historical comparability.

Power Usage Effectiveness (PUE) analysis required understanding complex interactions between cooling system design, environmental conditions, and workload characteristics. I developed sophisticated approaches for using campus-specific disclosed values where available whilst implementing peer-benchmarked estimates for non-disclosing providers, representing a pragmatic balance between analytical accuracy and comprehensive coverage.

I conducted comprehensive analysis of energy consumption patterns across different AI model architectures and deployment configurations, requiring detailed understanding of inference requirements and infrastructure optimisation strategies. OpenAI's model family analysis demonstrated substantial variation in per-query energy consumption, with GPT-3.5 Turbo achieving 0.00025 kWh per query (approximately £0.0001 in energy costs) whilst the o1-series requires 0.00390 kWh per query (approximately £0.0015 in energy costs) due to advanced reasoning capabilities requiring multiple inference passes (Microsoft Corporation, 2024).

Infrastructure provider comparison revealed distinct efficiency profiles with important implications for deployment strategy recommendations. Azure's integrated platform approach delivers superior efficiency for standard language model workloads, achieving 15-20% lower per-query consumption through shared infrastructure optimisation. CoreWeave's specialised infrastructure demonstrates significant advantages for sustained compute workloads, though often at premium pricing that affects total economic viability (CoreWeave Inc., 2025).

4.4 Validation and Quality Assurance

Implementing multiple validation approaches across different analytical frameworks proved essential for building confidence in results whilst identifying potential sources of systematic error. I developed comprehensive validation protocols that included capital expenditure analysis, GPU-hours calculation methods, and cross-provider benchmarking studies that provided independent estimates for consistency and accuracy assessment.

CoreWeave's detailed disclosure practices enabled comprehensive Track A analysis with confidence intervals within ±10%, validating the framework's ability to produce reliable results given appropriate transparency levels whilst demonstrating the critical importance of industry disclosure practices for analytical accuracy (CoreWeave Inc., 2025). This validation success highlighted both the potential accuracy achievable with high-quality data and significant limitations imposed by limited transparency practices across the industry.

Cross-provider comparison analysis revealed significant variations in energy efficiency that correlated strongly with disclosed operational practices and sustainability commitments. Google's industry-leading PUE of 1.09 and comprehensive renewable energy procurement strategy demonstrated achievable efficiency benchmarks whilst providing validation for the methodology's ability to identify meaningful performance differences across providers.

Economic validation involved comprehensive cross-referencing of energy consumption estimates with disclosed operational costs where available, providing independent verification of cost modelling assumptions whilst identifying potential optimisation opportunities. This analysis revealed that energy-efficient providers often achieve total cost advantages of 25-40% for inference-intensive workloads, making efficiency measurement critical for procurement decisions (Microsoft Corporation, 2024).

4.5 Stakeholder Engagement and Communication

Managing stakeholder relationships proved both challenging and educational, requiring development of sophisticated communication strategies that could address diverse information needs whilst maintaining analytical rigour. Limited direct management feedback during the analysis phase created challenges around ensuring alignment with strategic priorities, necessitating proactive approaches to validation through informal discussions with permanent team members.

I developed systematic approaches for maintaining stakeholder engagement through regular progress updates and methodology presentations. These interactions helped identify potential analytical blind spots, validate methodological approaches, and ensure alignment with CCAF's broader research agenda whilst building my capabilities in managing complex stakeholder relationships under ambiguous guidance conditions.

The interdisciplinary nature of the target audience required developing multiple presentation formats optimised for different stakeholder groups. Executive summaries focused on strategic implications and economic considerations that directly supported business decision-making processes, whilst technical appendices provided detailed methodology documentation for academic audiences requiring deeper analytical validation.

Regular discussions with CCAF team members proved invaluable for methodology validation and results interpretation. These conversations leveraged the team's diverse expertise spanning finance, technology, and policy whilst providing crucial perspectives on practical applications and implementation challenges that enhanced the strategic relevance of analytical outputs.

Industry engagement opportunities revealed strong interest in standardised approaches to AI energy measurement, particularly given growing regulatory requirements and cost pressures affecting technology procurement decisions. Feedback from financial sector stakeholders consistently emphasised the importance of connecting energy efficiency metrics to total cost of ownership calculations, highlighting the economic relevance of environmental performance optimisation.

5. Considerations for the Future

5.1 Methodological Enhancement Opportunities

Reflecting on the project experience, I identify several critical areas where alternative approaches could have significantly enhanced accuracy and utility whilst building upon successful methodological foundations. Future iterations should prioritise integration of real-time telemetry data through API connections with cloud providers and monitoring services, enabling dynamic analysis of utilisation patterns and specific workload impacts on energy consumption whilst providing more accurate cost projections for operational planning.

This real-time integration capability would transform the analysis from static assessment to dynamic monitoring system capable of tracking industry evolution and supporting ongoing optimisation of AI deployment strategies. The technical infrastructure required would represent a significant investment but could provide substantial ongoing value for CCAF's research agenda whilst establishing the centre as a leading source of current intelligence on AI infrastructure performance.

Implementing automated data collection systems represents another critical improvement opportunity that could scale the analysis whilst maintaining analytical currency. Web scraping frameworks could systematically monitor SEC filings and sustainability reports for relevant updates, whilst API integrations could provide more granular utilisation data that would enable more sophisticated analysis and accurate strategic recommendations.

Advanced uncertainty quantification through Monte Carlo simulation techniques could provide more sophisticated confidence intervals whilst better communicating inherent challenges in analysing an industry characterised by limited transparency and rapid technological evolution. This approach would support more nuanced risk assessment for organisations making infrastructure decisions based on environmental performance criteria.

The carbon intensity integration discussed in the methodology framework deserves prioritisation in future development cycles, as it would transform the analysis from purely energy-focused to comprehensive environmental impact assessment. Regional grid carbon intensity varies dramatically and changes dynamically based on renewable energy availability and seasonal demand

patterns, creating opportunities for sophisticated optimisation strategies that consider both cost and environmental objectives simultaneously.

Economic forecasting capabilities should be integrated to project future cost trends based on technology development trajectories, energy pricing evolution, and regulatory changes. This would transform the analysis from static assessment tool to dynamic strategic planning capability that could support multi-year infrastructure strategy development whilst addressing rapid evolution of AI economics.

5.2 Integration with MBA Learning

The project has effectively integrated multiple theoretical frameworks from the MBA programme whilst demonstrating their practical application in complex analytical and strategic contexts. Operations management principles proved directly applicable to understanding efficiency optimisation strategies, capacity planning challenges, and technology selection criteria that affect both cost performance and environmental outcomes across different provider types.

Strategic management frameworks provided essential structure for analysing competitive dynamics, differentiation strategies, and value creation mechanisms within the AI infrastructure market whilst connecting technical performance characteristics to broader business strategy considerations. These theoretical foundations enhanced analytical sophistication whilst ensuring that technical analysis translated into actionable strategic insights relevant for business decision-making.

Financial analysis methodologies from the MBA programme proved essential for developing sophisticated cost modelling approaches that could account for complex pricing structures and dynamic technology performance characteristics whilst connecting energy efficiency metrics to total cost of ownership calculations. This integration demonstrated the practical value of rigorous financial analysis training whilst highlighting opportunities for enhanced analytical sophistication.

Sustainability and stakeholder management frameworks provided crucial context for understanding regulatory requirements and policy implications that increasingly influence technology procurement decisions across the financial services sector. These theoretical perspectives enhanced strategic relevance whilst demonstrating the importance of interdisciplinary approaches to complex business challenges.

5.3 Professional Development Impact

This experience has significantly advanced my understanding of how to bridge technical depth with business applications, directly supporting career objectives in technology strategy consulting whilst providing practical experience in emerging technology assessment and business strategy development. The

ability to translate complex technical concepts into strategic insights whilst maintaining analytical accuracy represents a core competency for executive roles within technology-intensive industries.

The project reinforced my appreciation for interdisciplinary approaches to complex business challenges whilst highlighting the importance of rigorous methodology development in influencing industry practices and policy frameworks. I have developed valuable experience in handling complex, multi-source datasets whilst maintaining analytical rigour, skills that prove directly applicable to investment analysis, due diligence processes, and strategic planning activities.

Understanding the economic implications of energy efficiency in AI systems provides crucial insight into operational cost structures and competitive dynamics affecting technology companies and their clients. This knowledge proves directly relevant to investment analysis, strategic planning, and operational optimisation activities whilst addressing growing importance of sustainability considerations in business strategy development.

The experience of developing measurement methodologies for emerging technology areas provides valuable background for roles in venture capital, private equity, and strategic consulting focused on technology investments. Understanding how to quantify operational characteristics and efficiency differences across technology providers supports informed investment decision-making and portfolio company optimisation whilst addressing growing investor interest in environmental, social, and governance factors.

Future career development will build upon this foundation through additional exposure to emerging technologies, sustainability measurement frameworks, and regulatory development processes. The combination of technical depth, analytical capability, and business application provides a strong platform for contributing to the evolving intersection of technology and sustainable finance whilst addressing growing demand for professionals capable of navigating complex technical and economic challenges.

5.4 Long-term Strategic Impact

The project's outputs position CCAF to influence industry standardisation efforts and regulatory framework development around AI energy measurement whilst building upon credibility established through previous cryptocurrency analysis. Future engagement with standards bodies, regulatory agencies, and industry associations could help establish consistent disclosure frameworks that benefit all stakeholders whilst improving analytical accuracy and reducing compliance costs.

This standardisation influence represents a significant opportunity for positive impact that extends beyond academic research to practical improvements in industry transparency and sustainability performance across the rapidly growing AI infrastructure sector. CCAF's reputation for rigorous, evidence-based

analysis positions the centre uniquely to contribute to policy development whilst maintaining analytical independence and credibility.

The methodology development and analytical frameworks created through this project provide foundations for ongoing research initiatives that could address related challenges in technology assessment and sustainability measurement across multiple emerging technology domains. This work establishes CCAF's capabilities in technology infrastructure analysis whilst creating opportunities for expanded research programmes.

The experience reinforces the strategic value of academic-industry partnerships in addressing complex business challenges whilst highlighting opportunities for continued collaboration between research institutions and industry practitioners. I intend to maintain connections with CCAF whilst pursuing career opportunities that enable continued contribution to evidence-based policy development and industry standardisation efforts in emerging technology domains.

This project has moved me significantly closer to my long-term goals in technology strategy consulting by providing practical experience in emerging technology assessment combined with business strategy development. The skills developed in quantifying operational characteristics across technology providers directly support strategic consulting capabilities whilst addressing growing client demand for sustainability-focused analysis.

6. Conclusion

This CCAF placement successfully advanced my capabilities in bridging technical analysis with strategic business applications whilst addressing critical challenges in alternative finance. Developing sophisticated AI energy measurement methodologies demonstrates practical application of MBA theoretical frameworks to real-world problems with significant economic and environmental implications. Through rigorous data collection, validation, and stakeholder engagement, I developed advanced analytical capabilities that directly support my technology strategy consulting objectives.

By integrating top-down capacity analysis with granular, bottom-up approaches, I provided frameworks that inform strategic decisions around cost optimisation, sustainability reporting, and technology procurement. The project's outcomes position both CCAF and myself to influence industry standardisation efforts whilst contributing to evidence-based policy development in AI sustainability.

This interdisciplinary approach combined technical depth with business strategy insights exemplifies analytical sophistication expected from MBA graduates in technology-intensive sectors. The project experience strengthened my understanding of how emerging technologies reshape business operations whilst providing practical frameworks for evaluating complex technology investment decisions based on cost efficiency and sustainability considerations.

7. Appendix

The project culminated in building a professional online dashboard integrating all insights, data and visualisations. As sole author, I developed this dashboard for the CCAF to summarise trends and content from this individual project. The dashboard is accessible under the provided link.

https://aa-partner.shinyapps.io/ccaf_ai_energy_dashboard/

Username: ccaf
Password: g3n4l_01

Note: Login details should remain confidential, but assessors are welcome to explore this dashboard.

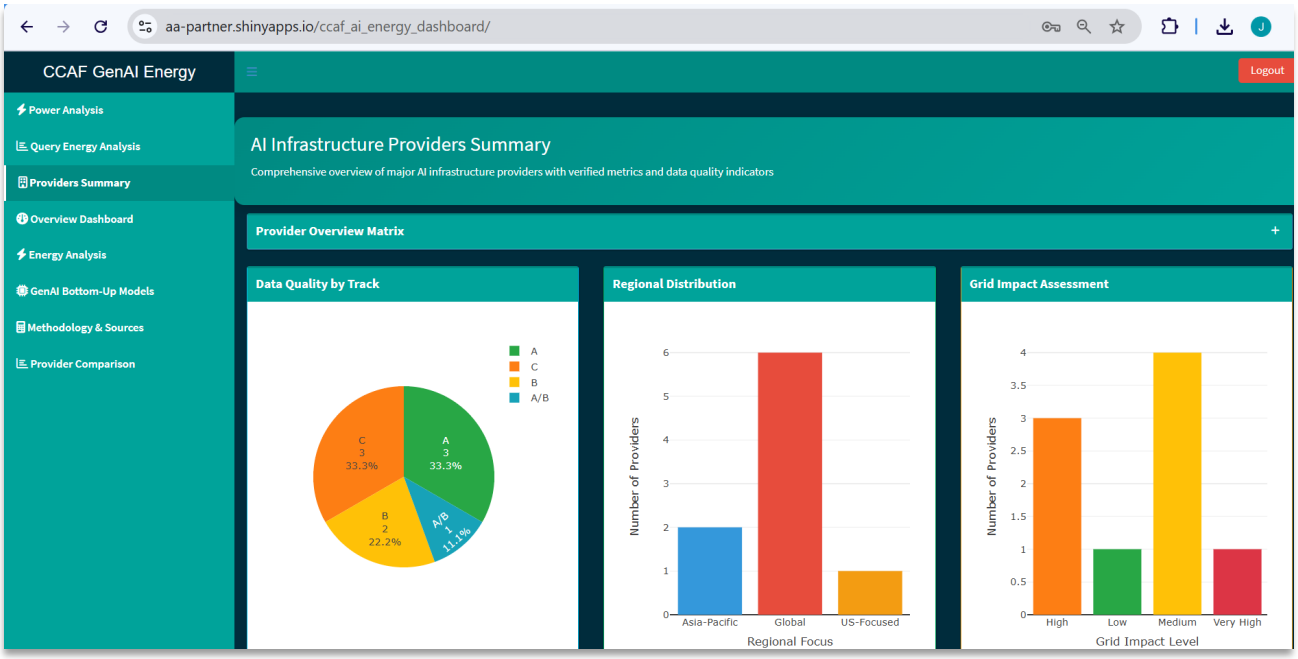


Figure 7.1 –Online Dashboard for GenAI Energy Assessment – Trends and Statistics

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